

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Monish Kumar K P (192312078)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Monish Kumar K P

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

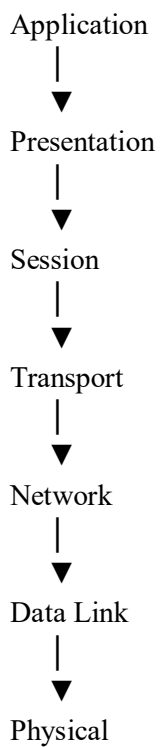
Analyze the problem and explain how the concept map helps in troubleshooting.

Solution:

ANNEXURE A – CONCEPT MAPPING

1. Complete the Concept Map

Diagram:



Missing Layers:

- Session Layer

- Data Link Layer

Analysis:

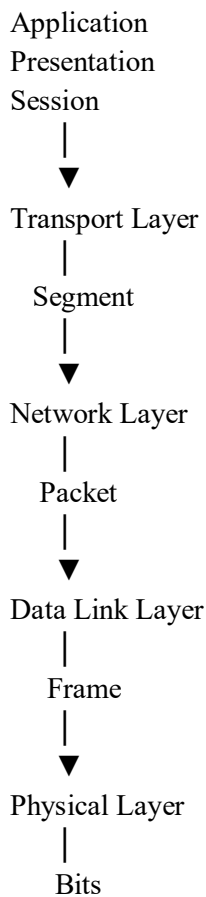
- Session Layer establishes, manages, synchronizes, and terminates communication sessions between applications. It also provides synchronization checkpoints to recover communication after failures.
- Data Link Layer converts packets into frames, provides MAC addressing, performs error detection using CRC, and controls access to the physical medium.

Explanation:

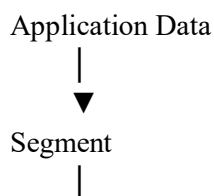
The Session Layer maintains continuous communication between devices, while the Data Link Layer ensures reliable frame transmission with error detection. Together, they provide reliable communication.

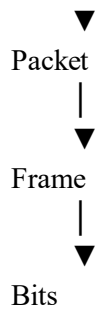
2. Concept Map: Data Units and OSI Layers

Diagram:



Analysis:





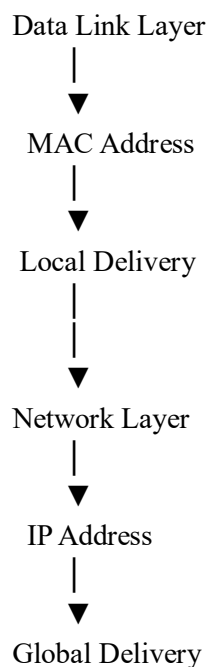
Explanation:

- Transport Layer → Segment
- Network Layer → Packet
- Data Link Layer → Frame
- Physical Layer → Bits

During transmission, application data is encapsulated into Segments, Packets, Frames, and finally Bits. At the destination, the reverse process (decapsulation) reconstructs the original data.

3. Concept Map: MAC Address and IP Address

Diagram:



Analysis:

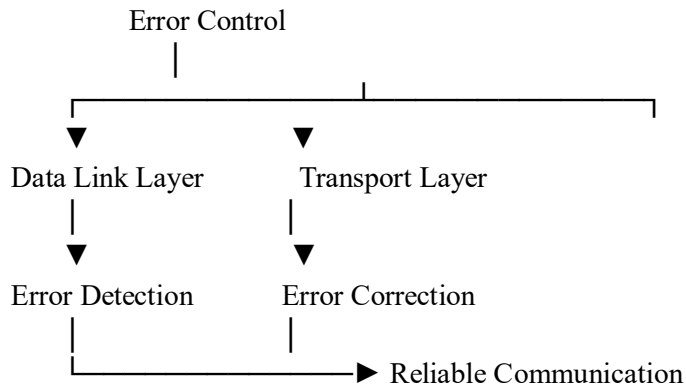
- MAC Address identifies devices within the same LAN.
- IP Address identifies devices across different networks.

Explanation:

MAC addresses ensure local communication, while IP addresses enable communication across multiple networks. Together they provide complete end-to-end data delivery.

4. Concept Map: Error Control

Diagram:



Analysis:

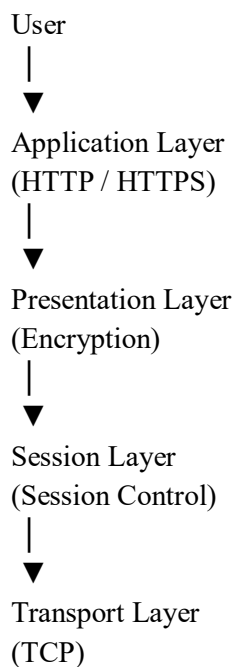
- Data Link Layer detects transmission errors using CRC.
- Transport Layer corrects errors through acknowledgements, sequence numbers, and retransmissions.

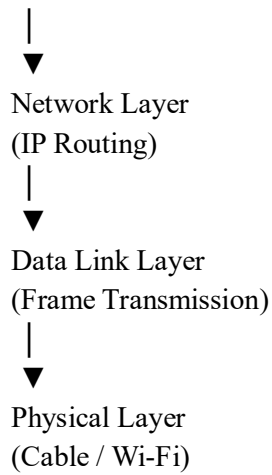
Explanation:

The Data Link Layer provides hop-by-hop reliability, while the Transport Layer provides end-to-end reliability. Together they reduce data loss and improve communication.

5. Concept Map: Web Browsing Using OSI Layers

Diagram:

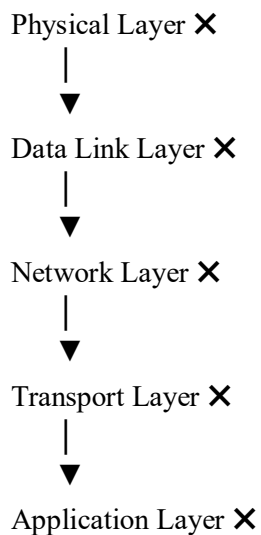




Analysis:

- Application Layer provides web services.
- Presentation Layer encrypts data.
- Session Layer manages communication.
- Transport Layer ensures reliable delivery.
- Network Layer routes packets.
- Data Link Layer frames data.
- Physical Layer transmits signals.

Effect of Layer Failure:

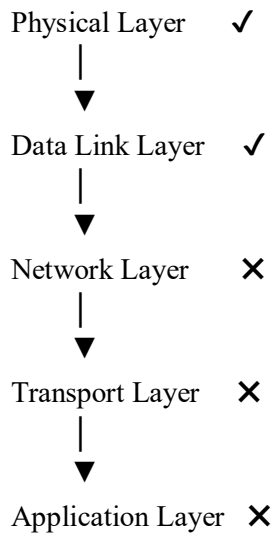


Explanation:

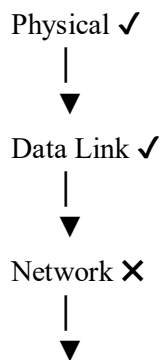
Each OSI layer depends on the lower layer. If one layer fails, the communication process is interrupted.

6. Network Troubleshooting Concept Map

Diagram:



Troubleshooting Flow:



Check:

- IP Address
- Subnet Mask
- Default Gateway
- Router
- Routing Table

Analysis:

- Physical and Data Link Layers are working correctly.
- Failure starts at the Network Layer.
- Because the Network Layer fails, the Transport and Application Layers cannot communicate.

Explanation:

Concept mapping helps identify the exact layer where communication fails. This layered troubleshooting approach reduces diagnosis time and helps network administrators solve problems quickly.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly

Explanation & Clarity	15%	Provides a clear, concise, and wellstructured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation
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